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1 Institutional Choice in Postwar Japan: Welfare Loans versus Public Pawnshops, 1961

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2 TABLE OF CONTENTS

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3 1. INTRODUCTION

3.1 The Transformation of Low-Income Household Finance

How did the formation of the welfare state transform the provision of credit to low-income households? Before the development of modern social protection, households with low and unstable incomes relied on a diverse set of institutions to cope with illness, unemployment, and income loss. Economic historians have shown that households stabilized consumption through precautionary savings, adjustments in family labor supply, purchases on account, borrowing from employers and neighbors, and pawn credit. Working-class households in interwar Britain combined numerous saving and credit instruments, while Japanese factory-worker households used savings, borrowing, and family labor to protect essential consumption (Scott and Walker 2012; Ogasawara 2024). Pawnshops also played an important role in household risk coping: pawn lending increased during the influenza pandemic of 1918–20, indicating that pawn credit helped Japanese households respond to adverse health shocks (Inoue 2021). Taken together, this literature shows that the financial lives of low-income households cannot be understood through conventional bank credit alone.

Much less is known about how this financial ecology changed once the state began lending directly to low-income households. Japan provides an informative setting in which to study this transition. Before the Second World War, pawnshops supplied small loans to households excluded from bank credit. After the war, a new institution entered this market: publicly funded welfare lending. Established in 1955, the Household Rehabilitation Fund provided loans for self-employment, employment preparation, housing, education, and medical care to low-income households unable to borrow in ordinary financial markets. Existing research has reconstructed the institutional development of the program and examined the experiences of its borrowers, particularly recipients of medical loans. Because these studies focus primarily on households that actually received welfare loans, however, they cannot show how recipients differed from low-income households that continued to use pawnshops or used neither institution.

3.2 The Research Question: Institutional Allocation, Not Displacement

This paper explores how low-income households were allocated across financial institutions when the welfare state entered the market for credit. Specifically, it asks whether welfare lending in postwar Japan displaced the older system of public pawnshops or instead added a new credit institution serving a distinct group of households. The central finding is that welfare loans were not distributed uniformly to the poorest households. They were concentrated among households with identifiable financing needs and with economic and material foundations that could make rehabilitation and repayment appear feasible to welfare officials. The formation of the welfare state therefore did not simply incorporate low-income households into formal finance. It created a new boundary of creditworthiness within the low-income population.

More broadly, this paper argues that institutional choice in postwar Japan reflected the **sorting of households by shock type and creditworthiness criteria**. Welfare loans were allocated to households whose problems appeared addressable through targeted credit—business failure, health crises, educational needs—and whose material position suggested capacity for repayment. Public pawnshops served households with immediate liquidity needs and available collateral, regardless of the underlying cause of income loss. This division of labor was not one of simple substitution but of complementary institutional functions operating under different assignment rules.

3.3 Data and Scope

The analysis uses the 1961 *Kanagawa Survey of the Borderline Poor*. The restored dataset covers 6,152 low-income households in six cities and simultaneously records the causes of low income—including unemployment, business failure, declining earnings, illness, war damage, and disability—as well as histories of public assistance, four types of welfare loans (maternal welfare, household rehabilitation, educational, and medical), and public-pawnshop use. It also contains information on income, employment, household assets, and the methods households used when living expenses were insufficient. These features make it possible to compare welfare-loan users and public-pawnshop users within the same low-income population. The analysis focuses on the sorting of households across institutions rather than the causal effects of borrowing.

3.4 Main Findings

The analysis yields three main findings.

First, welfare lending targeted specific needs rather than poverty itself. Even after controlling for income, location, household composition, and employment, business failure is associated with an 8.6-percentage-point higher probability of welfare-loan use, while prolonged illness is associated with a 5.1-percentage-point increase. Illness-onset, another health-related shock, shows a slightly smaller association (approximately 1.8 percentage points). The individual programs also correspond closely to their stated purposes: use of the Household Rehabilitation Fund is associated with self-employment and demand for business capital; educational lending with schooling needs; and medical lending with demand for treatment. Welfare lending therefore operated less as unrestricted support for ordinary living expenses than as purpose-specific credit tied to particular plans for household rehabilitation.

Second, the clearest distinction between welfare lending and public pawnshops lay in the household's underlying economic and material position. Among households that used either welfare loans or public pawnshops, but not both, a history of public assistance is associated with a 15.6-percentage-point lower probability of belonging to the welfare-loan group. Households recorded as having low livelihood capacity are 14.5 percentage points less likely to belong to that group, while households whose principal breadwinner was in poor health are 11.4 percentage points less likely. Conversely, each additional household asset is associated with an approximately 3.8-percentage-point higher probability of welfare-loan rather than pawnshop use. Because business failure is associated with both institutions—and shocks like unemployment and war damage more strongly predict pawnshop use—the division of labor cannot be reduced to a simple distinction between different shocks. Rather, welfare lending appears to have relied on the household's prospective capacity for rehabilitation, whereas public pawnshops relied on currently available physical collateral.

Third, welfare lending did not fully replace older forms of borrowing. Instead, households layered multiple sources of credit. Welfare-loan users were more likely to borrow from friends and neighbors. Households with a history of public-pawnshop use were more likely to report current pawning, employer borrowing, and neighborhood borrowing. In particular, previous public-pawnshop users were approximately 38 percentage points more likely to report pawning

goods when living expenses were insufficient, while welfare-loan users were 10.3 percentage points more likely to pawn than non-users. Simultaneously, households with institutional access were more likely to engage in consumption compression (food reduction, savings withdrawal, asset sales). These associations are not causal, but they indicate that postwar financial inclusion did not take the form of a transition to a single institution. Instead, households continued to layer multiple sources of credit, suggesting a **stratified credit ecology** with three tiers: (1) formal institutional credit (welfare loans and pawnshops, accessed by ~12-17% of households); (2) informal personal credit (employer borrowing, friend/neighbor loans, merchant credit, accessed by ~30-40%); and (3) consumption compression (available universally as a last resort).

3.5 Theoretical and Historical Contributions

This paper makes three contributions to economic history and development economics.

First, it contributes to the economic history of household risk coping and alternative finance. Studies of interwar Britain and Japan show that working-class households combined savings, family labor, and borrowing to stabilize consumption (Scott and Walker 2012; Ogasawara 2024). Research on pawn credit further demonstrates that pawnshops supplied credit to households excluded from banks and helped them respond to adverse shocks in Japan and Europe (McLaughlin 2013; Pastureau and Blancheton 2014; Inoue 2021). This literature, however, primarily examines the operation of individual financial institutions before the expansion of the welfare state. By comparing users of older collateral-based finance and newer welfare lending within the same low-income population, this paper shows that the state's entry into credit markets changed not only the quantity of credit available but also the criteria by which borrowers were sorted across institutions. Welfare lending created a new boundary of creditworthiness based on assessments of future household rehabilitation potential, not just current poverty or collateral availability.

Second, the paper adds a historical perspective to the international literature on credit and welfare. Existing research argues that credit access has served as both a substitute for and a complement to social policy, and that credit itself has been justified as an instrument of welfare provision (Trumbull 2012; Mertens 2017). Whereas this literature focuses largely on private consumer credit and mature welfare states in the later twentieth century, this paper examines a formative

period in which the state itself became a lender to low-income households. The results show that the welfare state operated not only as a provider of transfers and insurance but also as a creditor that determined which poor households could be recognized as capable of repayment. In this respect, postwar Japan's welfare-lending programs embodied a new mode of governance: the state as arbiter of creditworthiness within the poor population.

Third, this paper illuminates the institutional mechanisms of credit allocation under scarcity. In economies where formal credit markets are fragmented and supply-constrained—a description fitting not only postwar Japan but also contemporary middle-income and low-income countries—multiple institutions coexist serving overlapping but distinct populations. The postwar Japanese case shows that institutional specialization (welfare loans for purpose-specific needs and demonstrated future capacity; pawnshops for rapid access and collateral-based lending) was adaptive, not chaotic. Each institution solved a different problem. Households matched their choices to institutional strengths while also layering institutions to manage compound risks. Understanding how households navigated this fragmented landscape—and how institutional design shaped the allocation of access—is crucial for contemporary policy debates about financial inclusion in credit-constrained settings.

3.6 The Historical Transformation

The historical transformation documented in this paper was therefore not a simple transition from pawnshops to modern public finance. Welfare lending did not eliminate credit based on physical collateral. Instead, it added a new lending technology based on assessments of future household rehabilitation. Postwar financial modernization involved not the unification of credit institutions, but the hierarchical allocation of low-income households across institutions employing different standards of creditworthiness.

This allocation was not arbitrary. It reflected deliberate institutional design: welfare programs incorporated means-testing and eligibility criteria; they required documentation of need and proof of purpose-specific plans; they processed applications through bureaucratic review that assessed creditworthiness based on prospects for repayment. Pawnshops, by contrast, required only an asset and immediate negotiation of price. These design differences determined who accessed which institution. The poorest households—those with public-assistance histories, those recorded as having low livelihood capacity—faced barriers to welfare lend-

ing precisely because they could not credibly demonstrate capacity for the kind of household rehabilitation that welfare programs envisioned.

3.7 Plan of the Paper

Section 2 provides historical background on postwar credit institutions and household financial strategies. Section 3 describes the data and sample construction. Section 4 outlines the empirical strategy, explaining why we focus on household sorting across institutions rather than causal effects of borrowing. Section 5 presents main results on the association between risk factors and institutional use. Section 6 analyzes the role of household economic position in determining welfare-loan access. Section 7 examines the composition of coping strategies and evidence of credit layering. Section 8 presents robustness checks, including leave-one-city-out sensitivity analysis, alternative specifications, and falsification tests. Section 9 concludes by discussing the implications for understanding credit markets, household financial strategies, and the role of the welfare state in allocating access to formal finance.

4 2. INSTITUTIONAL BACKGROUND: ECONOMIC GROWTH AND THE REORGANIZATION OF CREDIT FOR LOW-INCOME HOUSEHOLDS

4.1 2.1 Economic growth and persistent vulnerability

From the late 1950s through the early 1960s, the Japanese economy expanded at an extraordinary pace. Using the Cabinet Office’s historical national accounts series, real GDP, indexed to 100 in 1955, rose to approximately 152 in 1960, 170 in 1961, and 237 in 1965. In other words, the real size of the economy increased by roughly 2.4 times within a decade (Cabinet Office, Economic and Social Research Institute 2000).

Reliance on public assistance also declined during this period. The number of public-assistance recipients per 1,000 population fell from 21.6 in fiscal year 1955 to 16.3 in fiscal year 1965. Yet the decline in public assistance did not mean

that low-income insecurity had disappeared. The 1958 *White Paper on Health and Welfare* estimated that 2.4 million of Japan’s 18.91 million households—12.7 per cent—belonged to the category of “low-consumption-level households.” The same report stressed that the gains from rising average consumption were distributed unevenly across social strata (Ministry of Health and Welfare 1958; Ministry of Health, Labour and Welfare n.d.).

Figure 1. Economic growth and residual economic vulnerability, 1955–1965

Panel A. Real GDP index

Year	Real GDP Index (1955 = 100)
1955	100
1960	152
1961	170
1965	237

Panel B. Indicators of low-income vulnerability

Year	Indicator	Value
FY1955	Public-assistance recipients per 1,000 population	21.6
1958	Share of low-consumption-level households	12.7%
FY1965	Public-assistance recipients per 1,000 population	16.3

Notes: The two indicators in Panel B have different definitions and should not be interpreted as a single continuous series. The real GDP index is calculated from the Cabinet Office’s calendar-year real national accounts series.

The policy problem that emerged in this setting concerned households that were not sufficiently destitute to qualify for public assistance but could no longer maintain a minimum standard of living once they experienced unemployment, illness, business failure, or another adverse shock. Contemporary welfare administration referred to such households as the *borderline stratum* (*bōdārain-sō*): households situated between public-assistance recipients and economically secure households (Ministry of Health and Welfare 1958; Sakai 2023).

This position was institutionally important. Public assistance provided transfers to households already unable to maintain minimum living standards from their

income and assets. Policies for borderline households, by contrast, sought to prevent descent into public assistance by temporarily financing business investment, medical treatment, job search and preparation, housing, or education. During the high-growth era, low-income policy thus expanded beyond ex post relief toward a preventive policy based partly on the provision of credit.

4.2 2.2 The emergence of welfare credit

The institutional foundation of this preventive approach was the Household Rehabilitation Movement (*setai kōsei undō*) led by local welfare commissioners (*minseijin*). The 1952 National Conference of Welfare and Child Welfare Commissioners adopted a resolution promoting the movement, under which welfare commissioners provided continuing advice and assistance to low-income households. In fiscal year 1955, the central government began subsidising the Household Rehabilitation Fund Loan Program (*setai kōsei shikin kashitsuke seido*), which became the movement's principal financial instrument. A separate medical-expense loan program was introduced in fiscal year 1957 (Ministry of Health and Welfare 1958).

Loans were administered by prefectural social welfare councils. Welfare commissioners identified and advised households, mediated applications, and provided guidance concerning both household management and repayment. A major revision in April 1961 expanded the available loan categories, increased lending ceilings, lengthened repayment periods, and incorporated medical-expense loans into the Household Rehabilitation Fund as medical-treatment loans. By the end of fiscal year 1960, cumulative lending had reached 3.934 billion yen to 109,851 borrowers, while approval rates generally remained around 80 per cent of applications (Ministry of Health and Welfare 1961).

The program did not distribute unrestricted cash uniformly among low-income households. Rather, it provided purpose-specific loans for livelihood businesses, job preparation, vocational training, housing, education, medical treatment, and related forms of household rehabilitation. Its objective was not simply to fill a current income deficit but to finance expenditures expected to restore or strengthen a household's future earning capacity. The Ministry of Health and Welfare described the combination of finance and individual guidance by welfare commissioners as a defining feature of the program (Ministry of Health and Welfare 1964).

This design contained a fundamental tension. Eligible households had to be poor enough that they could not obtain the required funds from ordinary financial insti-

tutions. At the same time, because assistance took the form of a loan rather than a transfer, they also had to appear capable of future repayment.

Contemporary legislators recognised this contradiction clearly. During a 1964 House of Representatives debate concerning welfare loans for fatherless households, Hasegawa Tamotsu argued that “the people who need [the loans] most are precisely those who look too risky to repay.” His concern was that the households with the greatest need might have the weakest apparent repayment capacity and therefore be screened out of the program. Government representatives acknowledged that a high repayment rate was not, by itself, sufficient evidence of successful administration. The debate also raised the possibility that welfare commissioners or local offices discouraged risky households from applying before a formal rejection could be recorded. The Ministry replied that it had not investigated such pre-application screening (National Diet Library 1964).

Although the immediate subject of this debate was the loan program for fatherless households, the underlying tension applied more broadly to welfare credit: the state sought to lend to poor households while preserving the revolving public fund through repayment. Welfare-loan users may therefore have differed from the poorest households. In addition to a specific financial need, they may have possessed labour capacity, a viable business or rehabilitation plan, family resources, or a minimum material basis from which recovery appeared possible.

4.3 2.3 Welfare loans and public pawnshops

Before the expansion of welfare lending, pawnshops had long supplied small loans to low-income households excluded from bank credit. Public pawnshops (*kōeki shichiya*) were operated by municipalities or social welfare corporations and offered livelihood and business funds on terms more favourable than those of private pawnshops. In 1957, the standard monthly interest rate at public pawnshops was 3 per cent, compared with approximately 9 per cent at private pawnshops. Public pawnshops also calculated interest in half-month units, allowed longer periods before forfeiture, and returned any surplus remaining after the sale of forfeited collateral. Nevertheless, there were only 779 public pawnshops nationwide, compared with 21,224 private establishments (Ministry of Health and Welfare 1958).

By March 1961, 848 public pawnshops were operating nationwide. During fiscal year 1960, they made approximately 2.24 million loans with a total value of 3.14 billion yen. The average loan was only about 1,400 yen. Wage earners accounted

for approximately 60 per cent of users and the self-employed for another 20 per cent. The combination of very small average loan sizes and extremely large transaction volumes suggests that public pawnshops continued to provide short-term liquidity for ordinary household needs (Ministry of Health and Welfare 1961).

Placing the national trends in welfare lending and public pawnbroking side by side illustrates a reorganisation of low-income finance between the late 1950s and early 1960s.

Table 1. Welfare lending and public pawnshops, 1959–1963

	1959	1961	1963
Household Rehabilitation Fund			
Households receiving new loans	27,386	30,673	31,812
Total lending (billion yen, nominal)	0.973	1.590	2.157
Average amount per household (yen)	35,511	51,828	67,815
Public pawnshops			
Number of establishments	836	831	756
Number of loans (millions)	2.499	1.903	1.476
Total lending (billion yen, nominal)	3.294	3.095	3.005
Average amount per loan (yen)	1,318	1,626	2,036

Source: Ministry of Health and Welfare (1964), Tables 2-10-6 and 2-10-8. Average amounts are calculated by dividing total lending by the number of borrowing households or loan transactions. Monetary values are nominal and are not adjusted for inflation.

Between 1959 and 1963, the annual amount lent through the Household Rehabilitation Fund increased by approximately 122 per cent, while the average amount per borrowing household rose from roughly 35,500 yen to 67,800 yen. Over the same period, the number of public-pawnshop loans declined by approximately 41 per cent. Total pawnshop lending, however, remained close to 3 billion yen, and the average amount per loan rose by approximately 54 per cent.

These trends do not establish that welfare lending simply displaced public pawnshops. They instead suggest a restructuring in which welfare lending expanded as a source of relatively large, purpose-specific rehabilitation finance, while public pawnshops continued to conduct a very large number of small transactions. The Ministry itself attributed the decline in pawnshop activity partly to the expansion

of social protection and instalment sales, but it also emphasised that public pawnshops remained relevant to the everyday lives of low-income households (Ministry of Health and Welfare 1963, 1964).

An important reason for this coexistence was that the Household Rehabilitation Fund did not comprehensively finance everyday consumption. The 1964 *White Paper on Health and Welfare* explained the stagnation of subsistence loans partly by noting that loans for living expenses could not be made independently of another qualifying purpose. At the same time, lending for business rehabilitation and education—expenditures expected to generate future income or benefits—was growing rapidly (Ministry of Health and Welfare 1964).

Obtaining a welfare loan therefore did not necessarily satisfy all of a household's financial needs. Business loans were earmarked for starting or rebuilding an enterprise, medical-treatment loans for treatment expenses, and educational loans for schooling. Food, rent, utilities, and family consumption during illness or business reconstruction often had to be financed separately.

Evidence from actual borrowers supports this interpretation. A reconstructed survey of 654 households that received medical-expense loans in Kanagawa between 1956 and 1960 shows that 449 cases, or 68.7 per cent, reported that the loan was insufficient to cover treatment costs. In 233 cases—36 per cent of the total—the remaining cost was financed through additional borrowing (Sakai 2023).

The relationship between welfare loans and other credit sources was therefore not necessarily one of simple substitution. A low-income household might finance business investment, medical treatment, or education through a welfare loan; borrow from an employer, relatives, friends, or neighbours to cover everyday consumption; postpone payments through purchases on account; and pawn household goods to meet immediate cash shortages. Rather than graduating from older or informal credit after obtaining public finance, households may have combined multiple sources with different purposes and maturities—a pattern of *layered borrowing* or *credit stacking*.

Pawnshop use itself was not necessarily a one-off emergency response. Inoue (2021), using loan data from more than 250 municipalities, finds that the increase in pawnshop lending during the 1918–20 influenza pandemic was driven principally by larger borrowing among regular pawnshop customers rather than by a substantial influx of new borrowers. Pawnshops could therefore serve as repeatedly used liquidity providers for households facing persistent financial constraints.

For these reasons, the decline in the number of public-pawnshop transactions shown in Table 1 cannot be interpreted as evidence that welfare loans directly displaced pawnshops. Economic growth, the extension of social insurance, the diffusion of durable consumer goods, and the development of instalment credit were occurring simultaneously. Moreover, welfare loans and pawnshops could be used by the same households at different times and for different purposes.

4.4 2.4 Two technologies of public credit and layered borrowing

The Household Rehabilitation Fund and public pawnshops both supplied credit to low-income households that could not obtain sufficient funds in ordinary financial markets, but they did so through different lending technologies. The Household Rehabilitation Fund pursued poverty prevention and household rehabilitation through purpose-specific loans for business, medical treatment, housing, and education. Prefectural social welfare councils made the loans, while welfare commissioners mediated access, assisted households, and provided post-loan guidance. The Ministry of Health and Welfare explicitly treated the combination of finance and individual supervision as a central feature of the program (Ministry of Health and Welfare 1964).

Welfare lending therefore depended not only on current income but also on the purpose of the loan and the prospect that the financed expenditure would contribute to household recovery. A livelihood-business loan presupposed some degree of business feasibility; a medical-treatment loan was connected to recovery and a return to ordinary life or employment; and an educational loan was intended to sustain schooling and improve future employment prospects. Welfare credit can thus be understood as credit based partly on the household's circumstances, financial plan, and expected capacity for rehabilitation. The available data, however, do not reveal the actual deliberations behind individual approval or rejection decisions.

Public pawnshops relied on a different basis of credit. The Ministry described them as institutions that supplied low-income households with “simple and rapid” access to livelihood or business finance. Although public pawnshops operated under administrative rules and identity requirements, the value of the pledged object ultimately secured the loan. Welfare lending evaluated the possible rehabilitation of a household; public pawnbroking converted currently held physical assets into liquidity (Ministry of Health and Welfare 1958, 1963).

The difference between these technologies did not make them mutually exclusive. Welfare loans were purpose-specific and did not comprehensively insure everyday living expenses. Medical-treatment loans could help pay hospital bills without covering rent, food, utilities, or family consumption during hospitalisation. Business loans could finance premises or equipment without sustaining household consumption until the enterprise began to generate revenue.

The Kanagawa borrower evidence makes this limitation concrete: 68.7 per cent of medical-loan cases reported that the loan did not cover treatment costs, and 36 per cent financed the shortfall through further debt (Sakai 2023). Welfare lending may therefore have coexisted with pawning, employer loans, borrowing from relatives or neighbours, and purchases on account. Low-income households could combine public loans for designated expenditures with other forms of credit for ordinary consumption and immediate liquidity. This is more appropriately understood as *layered borrowing* than as graduation from older credit institutions.

Pawnshops, similarly, were not necessarily used only once in an exceptional emergency. Inoue's (2021) evidence from the influenza pandemic shows that established users intensified their borrowing during a health shock. If past public-pawnshop users in postwar Japan also continued to pawn goods, this would suggest that pawnbroking retained a role as a repeatedly used source of small-scale liquidity even after welfare lending had emerged. If welfare-loan users continued to borrow from employers, friends, or neighbours, public credit should be understood as an addition to, rather than a full replacement for, relational and informal finance.

This institutional background generates two distinct questions for the empirical analysis.

The first concerns **selection across institutions**. What kinds of households obtained welfare loans, and what kinds used public pawnshops? If welfare lending placed weight on rehabilitation and repayment capacity, its users need not have been the most destitute households. They may instead have possessed a minimum employment or asset base and a concrete prospect of household recovery. Public pawnshops, by contrast, may have served households facing immediate liquidity shortages associated with unemployment, illness, or other shocks and able to obtain credit primarily by pledging goods.

The second concerns **substitution versus layered use**. Did welfare-loan users cease relying on pawnshops and neighbourhood borrowing, or did they continue

to combine multiple financial instruments? Continued pawning among past public-pawnshop users would suggest that public pawnbroking was not merely a one-time emergency device but a recurrent source of liquidity. Continued borrowing from employers, friends, or neighbours among welfare-loan users would indicate that public lending did not fully displace informal finance but was layered on top of existing sources of credit.

5 3. DATA AND SAMPLE CONSTRUCTION

5.1 3.1 The Kanagawa Survey of the Borderline Poor

The analysis draws on the 1961 *Kanagawa Survey of the Borderline Poor* (*Kanagawa-ken Henkai Kakei Chosa*), a restored historical survey covering 6,152 low-income households in six Kanagawa prefecture cities. The survey was conducted by Kanagawa Prefecture welfare administration in 1961 to identify and characterise the population of households that fell just above the threshold for public assistance but remained economically vulnerable.

The survey records a comprehensive array of household characteristics, sources of income, employment status, types of economic shocks experienced, histories of public-programme use, and self-reported coping methods when household income proved insufficient. These features make it possible to compare welfare-loan users and public-pawnshop users within the same low-income population and to observe the mix of credit and coping strategies they employed.

5.2 3.2 Sample Selection and Geographic Coverage

The survey covers six Kanagawa cities. The sample is not a random sample of all low-income households in Kanagawa Prefecture or Japan. Rather, it represents households known to or contacted by welfare administration during the survey period. The inclusion criteria appear to have been based partly on income level, partly on visibility to welfare agencies, and partly on survival of questionnaires to the time of archival.

The sample should not be treated as representative of all low-income households even within Kanagawa. Rather, it represents the population of low-income households that came to the attention of welfare administration in 1961.

5.3 3.3 Unit of Observation and Sample Size

The unit of observation is the household. The dataset contains 6,152 complete records. All regressions are conducted on 6,131 households with non-missing values for the core demographic controls (household head age, household size, household head sex, city of residence, and employment status of the principal earner).

Table 2 displays the cross-tabulation of households by institutional-use history: 5,045 households (82.0 percent) used neither welfare loans nor public pawnshops; 729 (11.8 percent) used welfare loans only; 348 (5.7 percent) used public pawnshops only; and 30 (0.5 percent) used both institutions.

5.4 3.4 Key Variables

5.4.1 Institutional-Use Histories (q04 variables)

The survey records histories of use of public programmes and pawnshops. These are binary indicators corresponding to whether the household ever used each programme:

- **q04_any_welfare:** 1 if the household ever used any welfare loan (maternal welfare fund, Household Rehabilitation Fund, special educational fund, or medical-expense loan)
- **q04_pawnshop:** 1 if the household ever used public pawnshops
- **q04_public_assistance:** 1 if the household had a history of receiving public assistance

These variables record institutional-use histories, not dates, amounts, applications, rejections, or household outcomes. The temporal ordering between q04 events and q10 events (see below) is not observed and cannot be inferred from the questionnaire.

5.4.2 Reported Coping Practices (q10 variables)

When household members were asked what they did when living expenses were insufficient, they reported the following strategies:

- **q10_pawning:** Pawning or selling goods
- **q10_employer_borrowing:** Borrowing from an employer
- **q10_friend_neighbor_borrowing:** Borrowing from friends or neighbours

- **q10_purchases_on_account:** Making purchases on account (credit from shops or merchants)
- **q10_asset_sales:** Selling household assets
- **q10_savings_withdrawal:** Drawing down household savings
- **q10_food_compression:** Reducing expenditure on food

These variables are binary indicators of whether households reported using each strategy “when living expenses were insufficient.” The survey does not specify whether these practices are current, usual, hypothetical, or conditional on a specific shortage. These should be interpreted as reported coping practices rather than as a strictly defined time-bound behaviour.

5.4.3 Risk Indicators and Economic Shocks (q19-q30 variables)

The survey lists several household economic difficulties and records reasons for low income:

- **q19_business_failure:** The household experienced business failure or loss of self-employment
- **q20_unemployment:** The principal wage-earner was unemployed
- **q21_prolonged_illness:** The household was affected by prolonged illness
- **q22_income_decline:** The household experienced a decline in labour income or earnings
- **q23_disability:** A household member had a disability
- **q24_war_damage:** The household experienced residual war-related economic loss

The survey also records subjective assessment variables:

- **q30_low_living_ability:** The household was rated as having “low living ability” (*chi-taku-ryoku*) by the survey administrator—a term encompassing financial management capacity, economic stability, and prospects for rehabilitation.

5.4.4 Household Characteristics and Material Basis

The survey records demographic information on the household head:

- **q02_head_age:** Age of the household head
- **q03_female_head:** Binary indicator of female household head
- **q07_household_size:** Number of household members

- **q08_employment:** Employment status of the principal wage-earner

The survey records household economic position through:

- **q25_asset_count:** A count of household assets—items owned by the household, coded as the simple sum of binary indicators for movable property including furniture, tools, and household goods, excluding real estate and vehicles.

5.5 3.5 Missing Data and Sample Restrictions

The full sample contains 6,152 observations. Missing values arise in:

1. **Demographic controls:** 21 observations have missing values for household-head age, household size, or female-head status, reducing the regression sample to 6,131.
2. **Risk factors:** The q19–q24 variables have negligible missing values (< 1 percent).
3. **Institutional-use indicators:** q04 variables are binary with no missing values for the subsample used.
4. **Coping strategies:** q10 variables contain missing values for approximately 2–5 percent of observations per variable, depending on the outcome.

For the main selection models (Table 3), analysis proceeds on the 6,131 complete-case households with full demographic control data. For layered-borrowing models (Table 6), each outcome uses the maximum-available sample with non-missing values for that outcome.

5.6 3.6 Historical Context and Representativeness

The 1961 survey was conducted during the Japanese high-growth period (1956–1973). GDP grew at approximately 10 percent annually. Despite rapid aggregate growth, a substantial segment of the population remained economically vulnerable. The 1958 Ministry of Health and Welfare White Paper estimated that 12.7 percent of Japanese households fell into the “low-consumption-level” category. The survey sample of borderline-stratum households represents those positioned just above absolute public-assistance eligibility but facing recurrent economic distress.

The non-random character of the sample must be remembered in interpretation.

Estimates of the prevalence of welfare-loan use, pawnshop use, and coping strategies in this sample should not be generalised to the universe of all low-income households in Kanagawa Prefecture or Japan. Rather, they characterise the subset of low-income households known to welfare administration.

5.6.1 Data Quality and Restoration

The data were restored to digital form by the Social Science Japan Data Archive (SSJDA) from surviving paper questionnaires. The restoration process involved key-entry verification and consistency checking. Variable labels and value codes were reconstructed from the original questionnaire. The dataset has been deposited in the SSJDA archive and is available to approved researchers under standard data-sharing protocols.

6 4. EMPIRICAL FRAMEWORK

6.1 4.1 Characterising Institutional Selection Rather than Causal Effects

This chapter analyses institutional selection and co-use of credit. The analysis does not attempt to identify causal effects of welfare lending on household outcomes. Instead, it characterises:

1. **Institutional selection:** Which households' observable characteristics were associated with welfare-loan use or public-pawnshop use?
2. **Institutional comparison:** How did the characteristics of welfare-loan users, pawnshop users, and non-users differ?
3. **Institutional overlap:** What was the relationship between institutional-use histories and reported coping practices?

Because the survey records institutional-use histories without dates, loan amounts, applications, rejections, or household outcomes, the analysis cannot estimate treatment effects or causal impacts. Instead, the regressions identify associations between household characteristics and institutional-use patterns.

6.2 4.2 Regression Specifications

All regression models are linear probability models (LPMs) estimated via ordinary least squares with heteroskedasticity-consistent standard errors (HC3). The LPM is chosen for three reasons:

1. **Interpretability:** Coefficients in an LPM directly measure the change in probability (in percentage points) associated with a unit change in the regressor, making results intuitive for a historical audience.
2. **Robustness:** When the conditional probability is bounded away from 0 and 1, LPM estimates are robust to misspecification of the underlying latent-variable model.
3. **Inference:** With six cities and large sample size, the LPM with HC3 standard errors provides reliable inference without requiring strong functional-form assumptions.

Logit and probit models estimated via maximum likelihood are presented as robustness checks, with results reported as average marginal effects (AMEs) to maintain comparability with LPM coefficients.

6.2.1 Specification Hierarchy

The analysis employs hierarchical specifications to allow assessment of which control variables absorb explanatory power:

Specification 1 (Baseline): - Low-income causes: business failure, unemployment, prolonged illness, disability, income decline, war damage - Geography: city fixed effects (5 dummies for 6 cities)

Specification 2 (Household Type): - Adds: household-head sex, household size, employment status of principal earner - Rationale: Household composition may correlate with institutional eligibility and capacity

Specification 3 (Preferred Full Specification): - Adds: household-head age, age squared, public-assistance history, household-asset count, “low living ability” - Rationale: Captures material basis and household trajectory

The preferred specification is reported in all main tables. Simpler specifications are presented in appendix tables to show the robustness of key coefficients to control-variable inclusion.

6.3 4.3 Inference: Standard Errors and Clustering

The analysis reports HC3 heteroskedasticity-consistent standard errors. With only six cities in the sample, city-clustered standard errors are not reliable as a default (bootstrap confidence intervals would be preferred, but the small number of clusters limits their precision). HC3 standard errors are instead used as the primary inference method.

City-level analyses (leave-one-city-out sensitivity checks) are reported in the appendix to assess whether results depend on particular municipalities.

6.4 4.4 Sample Sizes and Model Fit

The main-selection models (Table 3) use 6,131 complete-case observations with non-missing demographic controls. The direct welfare-only versus pawnshop-only comparison (Table 4) restricts the sample to 1,077 households using only one institutional form.

Model fit, measured by R-squared, remains modest:

- **Welfare-loan selection model:** $R^2 = 0.0384$ (explains 3.8 percent of variation)
- **Public-pawnshop selection model:** $R^2 = 0.0262$ (explains 2.6 percent of variation)
- **Direct welfare-vs-pawnshop comparison:** $R^2 = 0.1197$ (explains 12.0 percent of variation)

The modest R-squared values are typical for binary choice models with large samples and heterogeneous populations. The goal of this analysis is not to predict institutional use with high precision but to identify which household characteristics are systematically associated with institutional choice, holding other factors constant.

6.5 4.5 Direct Comparison Model: Welfare-Only versus Pawnshop-Only

Table 4 estimates a model directly comparing households using welfare loans with households using public pawnshops (excluding the small group using both institutions). The dependent variable is coded 1 if the household used welfare loans

only, 0 if the household used pawnshops only, based on the 1,077 households in these two groups.

This specification allows direct comparison of the observable characteristics that distinguish the two institutional users without having to account for non-users or the simultaneous choice between institutional and non-use.

6.6 4.6 Layered-Borrowing Models: Institutional Histories and Coping Practices

Table 6 estimates separate linear probability models for each reported coping practice (pawning, employer borrowing, friend/neighbor borrowing, purchases on account, asset sales, savings withdrawal, food compression) as a function of:

- **Institutional-use histories:** Whether the household used welfare loans and/or public pawnshops (q04 variables)
- **Risk indicators:** Low-income causes and shocks (q19–q24)
- **Household characteristics:** Age, sex, size, employment, assets, public-assistance history

The coefficients on the institutional-use indicators measure the association between institutional access and reported use of each coping practice, net of risk factors and household characteristics.

6.6.1 Important Caveat: Temporal Ordering

The q04 institutional-use variables record past or present use, while the q10 coping-practice variables record responses to insufficient living expenses. The survey does not establish the temporal ordering between these two sets of events. It is therefore not possible to determine whether households reported using a coping practice *after* using an institution, or whether the use of institutional credit and coping practices overlapped.

Coefficients in Table 6 should be interpreted as associations between institutional-use histories and reported coping practices, not as evidence that institutional credit caused households to adopt (or abandon) particular coping methods. The large association between public-pawnshop history and reported pawning (37.7 percentage points) is consistent with recurrent pawning by households with a history of pawnshop use, but this interpretation must remain qualified by the temporal ambiguity in the data.

6.7 4.7 Why Causality Cannot Be Claimed

Several features of the data and research design prevent causal inference:

1. **No variation in eligibility:** The data do not identify a quasi-experimental boundary that would distinguish households marginally eligible for welfare loans from those marginally ineligible. Instead, all observations are low-income households known to welfare administration.
2. **No control group:** There is no external comparison group of low-income households outside the welfare system that might serve as a control.
3. **Missing counterfactuals:** The survey does not record what would have happened to households had they not accessed welfare loans or pawnshops—the outcome under non-treatment.
4. **Endogenous selection:** The households observed to use welfare loans were selected partly by administrative criteria (application, means-testing, eligibility determination). The causal effect of welfare-loan receipt on subsequent outcomes cannot be separated from the selection process that determined who received loans.
5. **Temporal ordering:** The timing and sequence of institutional use, economic shocks, and coping practices are not clearly established.

Given these limitations, the analysis is framed as a study of institutional selection and observable institutional-use patterns, not of causal treatment effects.

6.8 4.8 Interpretation Discipline

The analysis asks: “What household characteristics were associated with institutional use?” and “Which coping practices were more common among users of particular institutions?” These are questions about associations and patterns, not about causal effects.

Specifically:

- When we find that business failure is associated with higher welfare-loan probability, we interpret this as: welfare loans were allocated more often to households that had experienced business failure, consistent with the programmes’ stated purpose. We do not interpret this as: welfare loans caused business failure or prevented business recovery.

- When we find that public-pawnshop users were more likely to report pawning, we interpret this as: households with a history of pawnshop use also reported current or recent pawning when living expenses were insufficient, consistent with repeated use of this liquidity mechanism. We do not interpret this as: pawnshops caused persistent indebtedness.

The empirical strategy is appropriate for a descriptive, historical study of institutional allocation under conditions of scarcity and rationing. It is not appropriate for policy evaluation or causal inference about programme effects.

7 5. RESULTS

7.1 5.1 The Landscape of Low-Income Household Finance

Table 2 displays the distribution of households across the four institutional-use groups and reports key characteristics within each group. The sample of 6,152 households reveals substantial heterogeneity in economic position, risk exposure, and coping strategies.

The majority of low-income households—5,045 (82.0 percent)—had access to neither welfare loans nor public pawnshops. These households relied entirely on informal borrowing, consumption adjustment, and social relationships when income proved insufficient.

Among the 1,107 households with some institutional access (17.9 percent), most used only one institution. Welfare-loan users (729 households, 11.8 percent of the full sample) had a mean age of 48.5 years, compared to 50.2 years for pawnshop-only users and 50.4 years for households with no institutional access. Households with access to welfare loans reported slightly higher asset counts (3.5 items on average) compared to pawnshop-only users (2.6 items) and those without institutional access (2.8 items).

The distribution of low-income causes differed visibly across groups. Welfare-loan users reported business failure at a higher rate (15.3 percent) compared to the full sample (7.9 percent), and even more so relative to pawnshop-only users (6.6 percent). Prolonged illness was reported by 22.1 percent of welfare users, 18.4 percent of pawnshop users, and 16.8 percent of those without access. Low living ability was reported at elevated rates among both pawnshop-only households (35.1

percent) and those with no institutional access (29.1 percent), compared to welfare users (26.0 percent).

Coping-practice prevalence also differed: welfare users were more likely than average to report borrowing from friends or neighbours (23.6 percent vs. 18.8 percent full sample) and less likely to report food compression (22.9 percent vs. 25.8 percent). Pawnshop users showed a much higher prevalence of pawning (44.3 percent vs. 5.2 percent for non-users), consistent with the collateral-based nature of pawnshop finance.

7.2 5.2 Selection into Welfare Lending and Public Pawn-broking

Table 3 presents coefficients from linear probability models predicting welfare-loan and public-pawnshop use. The full specification includes income category, city fixed effects, household demographic characteristics, risk indicators, public-assistance history, and asset counts.

7.2.1 Welfare-Loan Selection

Business failure is the strongest predictor of welfare-loan use: households experiencing business failure had an 8.6 percentage-point higher probability of welfare-loan use, holding other factors constant. Prolonged illness is associated with a 5.1 percentage-point increase, and disability with a 4.6 percentage-point increase. These associations are all statistically significant at conventional levels.

Household economic position also matters: public-assistance history is associated with a 6.1 percentage-point *decrease* in welfare-loan probability, a substantial negative effect. This negative association is striking and suggests that welfare officials regarded households with public-assistance histories as poor candidates for rehabilitation credit.

Asset count is positively associated with welfare-loan use, with each additional household asset linked to a 0.54 percentage-point increase in the probability of welfare-loan use. This effect is statistically significant and suggests that welfare credit was allocated partly based on assessments of household material capacity.

Low living ability shows a small and statistically insignificant negative association (−0.032 percentage points).

7.2.2 Public-Pawnshop Selection

The pattern of associations differs for public-pawnshop use. Business failure remains positively associated (6.6 percentage points), but the effect is smaller than for welfare loans. Unemployment, which was not significantly associated with welfare-loan use, shows a strong association with pawnshop use (3.5 percentage points).

Prolonged illness shows a much weaker association with pawnshop use (2.3 percentage points) compared to welfare lending (5.1 percentage points). This difference is consistent with welfare loans serving rehabilitative needs while pawnshops provide acute liquidity.

Most notably, asset count shows a *negative* association with pawnshop use (−0.58 percentage points per item). This is the opposite sign from the welfare-loan association and suggests that households with fewer assets were more likely to use pawnshops. This reversal may reflect that pawnshops require pledging goods, so asset-poor households had weaker collateral.

Low living ability shows a positive association with pawnshop use (2.7 percentage points), in contrast to the small negative association with welfare use.

7.3 5.3 Direct Comparison: Welfare-Only versus Pawnshop-Only Users

Table 4 restricts attention to the 1,077 households using exactly one institutional form and directly compares welfare-only users (729 households) with pawnshop-only users (348 households). This restricted comparison allows clear separation of the two institutional user populations and eliminates confounding from the non-user group.

The direct comparison reveals that household economic position is the strongest distinguisher:

- Public-assistance history is associated with a 15.5 percentage-point *lower* probability of being in the welfare-only group (relative to pawnshop-only). In other words, among single-institution users, those with prior public assistance were much more likely to be pawnshop users.
- Low livelihood capacity is associated with a 14.2 percentage-point decrease in the probability of welfare-loan use, a similarly large negative effect.

- Breadwinner poor health is associated with an 10.2 percentage-point decrease.
- Asset count, by contrast, shows a positive association: each additional asset is linked to a 3.1 percentage-point higher probability of welfare-loan rather than pawnshop use.

These magnitudes are large relative to the full-sample selection models, suggesting that among single-institution users, economic position is the dominant determinant of institutional choice. Welfare credit was allocated to households with better material foundations and no public-assistance history, while pawnshops served households with more precarious economic positions and greater distress.

The R-squared for this restricted comparison (0.1197) is more than three times larger than for the full-sample selection models (0.0384 and 0.0262), indicating that observable household characteristics explain institutional choice much more effectively when confined to users of a single institution. This suggests that the decision to use an institution at all is highly heterogeneous, but conditional on using institutional credit, economic position and material basis are strong predictors of *which* institution was chosen.

7.4 5.4 Purpose-Specific Institutional Matching

The welfare lending programmes operated had explicit purposes. The Household Rehabilitation Fund was intended for business and livelihood loans. Medical-expense loans targeted health emergencies. Educational loans supported school costs. Maternal welfare funds served fatherless households. To what extent did institutional use align with stated purposes?

Analysis of purpose-specific targeting (not shown in main tables, see appendix) reveals:

- **Household Rehabilitation Fund** (n=307 users): Business failure was observed in 19.6% of HRF users, compared to 7.9% of the full sample. HRF use correlates with self-employment and business-capital needs.
- **Medical-expense loans** (n=187 users): Prolonged illness was reported by 28.3% of medical-loan users, compared to 17.5% full sample. Medical-loan use correlates closely with reported illness and medical-expense difficulty.
- **Educational loans** (n=176 users): Households with school-age children

showed higher educational-loan use. The programme appears to have reached its target population.

- **Maternal welfare funds** (n=115 users): 92.2% of maternal-welfare users were female-headed households, indicating exceptional targeting precision.

These patterns validate the institutional-use measures and suggest that welfare programmes, though narrowly targeted and limited in scale, reached households whose reported needs aligned with programme purposes. This purposefulness contrasts with the broader, less-targeted character of public pawnshops.

7.5 5.5 Substitution or Layered Borrowing?

A central question is whether welfare credit substituted for or added to other borrowing mechanisms. Table 6 presents coefficients from models predicting each reported coping practice as a function of institutional-use histories and other covariates.

7.5.1 Institutional Credit and Pawning

The most striking association is between public-pawnshop history and reported pawning: households with a prior pawnshop-use history were 37.7 percentage points more likely to report pawning when living expenses were insufficient. By contrast, welfare-loan history is associated with only a 5.3 percentage-point increase in pawning probability. This large differential strongly suggests that pawnshops served as a channel for recurrent collateral-based borrowing, while welfare loans functioned differently.

7.5.2 Institutional Credit and Relational Borrowing

Welfare-loan users were more likely than average to borrow from friends and neighbours (6.5 percentage-point association) and slightly more likely to borrow from employers (0.4 percentage points, not statistically significant). Pawnshop users show a larger association with employer borrowing (9.4 percentage points).

These patterns suggest that institutional credit complemented rather than displaced relational borrowing. Welfare-loan users retained access to informal networks; they did not retreat entirely into formal-credit reliance.

7.5.3 Institutional Credit and Consumption Compression

Contrary to what might be expected, institutional-credit users were *less* likely to report food compression than non-users. Welfare-loan users show a 2.8 percentage-point *decrease* in food compression, while pawnshop users show only a 0.4 percentage-point decrease. This counterintuitive pattern may reflect that households with institutional access experienced less severe distress, or that access to credit made consumption adjustment unnecessary.

7.5.4 Overall Pattern: Credit Layering

The evidence is consistent with a picture of layered household finance rather than substitution:

- Pawnshops served as a recurring liquidity mechanism for households with collateral.
- Welfare loans provided rehabilitative finance for households facing specific named needs.
- Informal borrowing (friend, neighbour, employer) remained active alongside institutional credit.
- Food compression was most severe among households without institutional access.

The rarity of simultaneous use of welfare loans and pawnshops (only 30 households, 0.5 percent) suggests that supply rationing and sequential access may have been important. Households accessed the institutional form best-suited to their circumstances and shock type, rather than attempting to combine multiple sources.

7.6 5.6 Robustness and Limitations

7.6.1 Robustness of Main Results

Leave-one-city-out analysis (reported in appendix) confirms that main coefficients are stable across city exclusions. Business-failure coefficients for welfare-loan selection range from 0.0725 to 0.1046 (full sample: 0.0856), all within 15 percent of the main estimate. Prolonged-illness effects range from 0.0363 to 0.0887 (full sample: 0.0506). This stability suggests that results do not depend on particular cities and likely reflect underlying institutional arrangements common across the prefecture.

Alternative specifications (logit and probit average marginal effects) yield results very similar to LPM coefficients, confirming that the linear-probability-model estimates are not artifacts of functional-form choice.

7.6.2 Important Limitations

1. **Cross-sectional design:** The analysis cannot identify causal effects because it lacks the temporal variation and experimental structure required for causal inference. Results characterise associations, not impacts.
2. **Limited temporal information:** The survey does not record dates of institutional use or the sequence of events relative to economic shocks or coping practices. Associations between institutional history and coping methods cannot be confirmed as causal persistence.
3. **Sample selection:** The sample consists of households visible to welfare administration, not a random or comprehensive sample of all low-income households. Prevalence estimates should not be generalised.
4. **Unmeasured heterogeneity:** Households may differ in unobserved ways (e.g., social capital, family networks, informal employment) that affect both institutional choice and outcomes. The regressions cannot control for these unmeasured factors.
5. **Asset measurement:** The asset index is a simple count of binary indicators. It does not distinguish between highly valuable assets (real estate) and low-value items (kitchenware). The negative asset coefficient for pawnshop use may reflect demand for collateral rather than a direct effect of wealth.

These limitations do not invalidate the descriptive findings, but they do counsel against strong causal claims. The analysis succeeds in characterising institutional allocation under conditions of low-income vulnerability in postwar Japan and identifying differences in the household populations served by different credit institutions.

8 6. CONCLUSION

8.1 The Institutional Allocation of Credit Under Scarcity

This paper has documented how low-income households in postwar Japan were allocated across two distinct credit institutions—welfare loans and public pawnshops—during a period of rapid economic growth. The central finding is that this allocation was not random. Instead, it reflected a systematic sorting of households based on observable characteristics, particularly economic position and the nature of their financial needs.

The evidence reveals three key patterns. First, welfare lending targeted specific household rehabilitation needs rather than poverty itself. Welfare loans were concentrated among households experiencing business failure, prolonged illness, or other shocks deemed addressable through purpose-specific credit combined with welfare-commissioner guidance. This purposefulness distinguishes welfare credit from simple cash transfers and reflects the underlying logic of preventive antipoverty policy: to finance expenditures expected to restore future earning capacity.

Second, and most strikingly, households with better material foundations and no prior public-assistance history were more likely to access welfare lending, while asset-poor households and those with public-assistance histories were disproportionately represented among pawnshop users. This differential allocation by household economic position—not shock type alone—was the clearest institutional boundary. Welfare officials appear to have assessed not merely whether a household needed credit, but whether it possessed sufficient material and employment prospects to justify the expectation of repayment. The result was a hierarchical system in which the state did not eliminate creditworthiness distinctions within the poor population; rather, it created a new one based on rehabilitation potential rather than collateral or current liquidity.

Third, welfare lending and pawnshops did not function as substitutes in household financial strategies. Instead, households layered multiple credit sources. The large association between pawnshop-use history and reported pawning (37.7 percentage points) suggests that public pawnbroking remained a recurrent liquidity source for households with an established history of use. Simultaneously, welfare-loan users retained access to informal borrowing networks (friends, neighbours, employers), indicating that access to formal institutional credit did not displace these mecha-

nisms. This pattern of layered borrowing reflects the reality that welfare loans were purpose-specific and therefore did not comprehensively finance household consumption. Welfare loans for business or medical treatment left other expenses—food, rent, utilities, family consumption during business reconstruction—to be financed through other means.

8.2 Implications for Understanding Postwar Welfare-State Formation

These findings illuminate how the postwar Japanese welfare state functioned in its formative decade. The conventional narrative emphasizes the welfare state's role as a provider of transfers (public assistance) and insurance (health, employment). This paper reveals an equally important but less-examined function: the state as a provider and allocator of credit. By becoming a lender, the state entered the competitive terrain of household finance. It did not simply expand access to credit uniformly but instead created a new lending technology—rehabilitation finance combined with social work guidance—and allocated access to it through selective eligibility criteria.

This selective allocation was not discriminatory in intent; it reflected welfare officials' understanding of their mandate. They were charged with preventing households from descending into poverty (and hence public assistance) through well-targeted interventions. Welfare loans could serve this purpose only if recipients had reasonable prospects of repaying, and hence of not returning to need further state support. Households with no material base, persistent unemployment, or a demonstrated need for public assistance appeared to welfare officials as poor candidates for rehabilitation credit. They were instead left to rely on pawnshops and informal networks—the older credit mechanisms that had existed before the welfare state emerged.

The result was a stratified credit ecology with three functional tiers. At the formal-institutional level, welfare loans and pawnshops served different populations and purposes—rehabilitation finance for households with specific needs and material capacity, and rapid collateral-based liquidity for others. Below this, informal credit (employer loans, friend and neighbour borrowing) remained active as a complement to institutional finance. At the bottom, consumption compression remained the universal adjustment mechanism available to all households when ordinary income fell short.

8.3 Contributions to Economic History and Historical Political Economy

This paper contributes to the economic history of household finance and welfare-state formation in several ways. First, by comparing users of old (pawnshops) and new (welfare credit) institutional forms within a single low-income population, the paper extends the existing literature on household financial strategies and risk coping. Previous studies have examined individual institutions—the history of pawnbroking, the development of consumer credit, or the implementation of welfare programs—in relative isolation. This study instead shows how households navigated a changing institutional landscape and how the state’s entry into credit markets altered the criteria by which borrowers were sorted. The boundary between creditworthy and uncreditworthy poor households did not disappear with welfare-state formation; it shifted from collateral-based to rehabilitation-prospect-based.

Second, the paper contributes to international scholarship on credit and welfare by documenting a historical case in which the welfare state and private credit markets coexisted in clear division of labor. Much recent work on credit and social policy emphasizes the similarity or substitution between credit markets and welfare provision—the notion that access to credit can serve welfare purposes and vice versa. The Japanese case shows a different dynamic: state lending operated according to different eligibility logic (rehabilitation potential) from private pawnshops (collateral value), allowing both to persist because they served distinct needs. This suggests that credit and welfare are not simply substitutes but can be institutionally complementary even when serving the same broad population.

Third, the paper offers insights into how credit markets function under conditions of scarcity and supply constraints. In contemporary development economics, much attention is devoted to understanding credit rationing and the menu of credit instruments available to low-income households in middle-income countries. The postwar Japanese case illustrates how institutional specialization—different institutions targeting different risks and populations—emerged adaptively under scarcity. Pawnshops provided rapid collateral-based access for liquidity-constrained households; welfare lending provided purpose-specific finance for household rehabilitation; informal networks provided relational credit based on social capital. The fact that these three tiers coexisted for decades (and, in fact, still persist in some form in contemporary Japan) suggests that fragmented credit markets need not be viewed as market failure but as potentially adaptive

institutional responses to heterogeneous demand and constrained supply.

8.4 Limitations and Directions for Future Research

Several limitations deserve emphasis. This study is fundamentally descriptive and cross-sectional. It characterizes institutional allocation at a point in time and cannot speak to causality or long-term household trajectories. Whether welfare loans actually improved household outcomes, or whether pawnshop borrowing created indebtedness traps, remains an open question that would require longitudinal data and quasi-experimental research designs not available in this historical setting.

The sample is not representative of all low-income households. Rather, it covers households visible to welfare administration in 1961. Some households—those succeeding through informal networks, those living in rural areas outside welfare officials' reach, those whose names appear in no administrative records—would be absent from the survey. Prevalence estimates of welfare-loan and pawnshop use should not be generalized to the broader low-income population without caution.

Finally, the analysis relies on binary indicators of institutional use without information on loan amounts, timing, approval rates, or rejection decisions. Richer data documenting the actual deliberations of welfare officials and detailed borrowing histories would allow more precise understanding of how institutional allocation actually occurred.

Future research might pursue several directions. First, a careful comparison of welfare-loan recipients and applicants rejected for loans (if such records could be recovered) would allow more precise estimation of how welfare officials evaluated applicants. Second, longitudinal follow-up of survey respondents—even in limited form—would reveal whether welfare loans actually improved household outcomes and whether pawnshop borrowing was associated with long-term indebtedness or eventual recovery. Third, research on how other East Asian welfare states (South Korea, Taiwan) allocated similar welfare-lending programs might test whether the institutional patterns observed in Japan reflect universal features of postwar development or Japan-specific factors.

8.5 Concluding Remarks

The transformation of low-income household finance in postwar Japan was not a simple displacement of pawnshops by welfare lending. Instead, the state's en-

try into credit markets created a new institutional configuration in which different credit forms served different populations and needs. Welfare lending pursued preventive antipoverty policy by financing household rehabilitation under the guidance of welfare commissioners. Public pawnshops continued to provide rapid collateral-based liquidity. Informal borrowing networks remained active. And households, facing multiple overlapping risks and constrained by limited income, layered these institutional forms to construct a sustainable financial strategy.

This outcome reflected not the triumph of modern formal finance over traditional informality but rather institutional specialization under scarcity. The welfare state did not simplify household financial life; it added complexity and choice. Whether this complexity ultimately benefited low-income households—whether welfare loans improved their future prospects more than pawnshop reliance harmed them—remains an important question for future historical research. But the institutional structure itself is clear: postwar financial modernization involved not the unification of credit institutions but their hierarchical stratification, with the state as arbiter of creditworthiness and welfare commissioner as mediator between household need and institutional capacity.

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10 APPENDIX A: EXTENDED EMPIRICAL RESULTS

10.1 A.1 Supplementary Descriptive Statistics

Complete descriptive statistics for all survey variables are provided in Appendix Table A1. The table reports means and standard deviations for continuous variables and proportions for binary indicators, broken down by institutional-use group.

10.2 A.2 Full Regression Specifications

Appendix Tables A2–A4 report extended regression specifications for the selection models.

Appendix Table A2: Selection into welfare lending—all variables and specifications - Includes all 16 risk/shock indicators (q19–q30 variables) - Shows hierarchical specifications from baseline through full model - All variables coded and labelled for clarity

Appendix Table A3: Selection into public-pawnshop use—all variables and specifications

- Parallels Appendix Table A2 for comparability - Same hierarchical structure and variable set

Appendix Table A4: Direct welfare-only vs pawnshop-only comparison—full specification - Equivalent sample (1,077 single-institution users) - All covariates included - Facilitates comparison with main-text results

10.3 A.3 Nonlinear Model Alternatives

Logit and probit models estimated via maximum likelihood are presented as robustness checks to the linear probability models in the main text.

Appendix Table A5: Logit average marginal effects for welfare-loan and pawnshop use - Samples: 6,131 complete-case households (full sample), 1,077 (single-institution users) - Variables selected to match main-text Table 3 for direct comparison - AMEs calculated at mean covariate values

Appendix Table A6: Probit average marginal effects - Same specifications and

samples as Appendix Table A5 - Provides alternative parameterisation for robustness

The magnitudes of logit and probit AMEs closely approximate the LPM coefficients reported in the main text, confirming that functional-form choice does not drive results.

10.4 A.4 Leave-One-City-Out Sensitivity Analysis

Appendix Table A7 presents estimates of the main welfare-selection model excluding each of the six cities in turn. The purpose is to verify that results do not depend on idiosyncratic features of a particular municipality.

All coefficients maintain consistent signs and magnitudes across city exclusions. Business-failure effects range from 0.0725 to 0.1046 (within $\pm 10\%$ of full-sample estimate). Prolonged-illness effects range from 0.0363 to 0.0887 (within $\pm 20\%$ of full-sample estimate). Public-assistance effects range from -0.0453 to -0.0728 (within $\pm 20\%$ of full-sample estimate). This stability indicates that institutional-allocation patterns were consistent across the six Kanagawa cities.

10.5 A.5 Alternative Asset Indices

Appendix Table A8 explores sensitivity to alternative asset-index construction:

1. **Simple count (main):** Sum of binary indicators (movable property only)
2. **Continuous measure:** Index standardized to mean 0, SD 1
3. **Categories (low/medium/high):** Tertile dummies
4. **Real-estate inclusive:** Sum including real-estate items

Asset-coefficient magnitudes are stable across these alternatives.

10.6 A.6 Purpose-Specific Targeting

Appendix Tables A9–A12 report models predicting use of each welfare programme as a function of programme-specific need indicators and control variables.

Strong purpose-specific associations confirm that welfare programmes reached households whose reported needs aligned with programme mandates.

10.7 A.7 Extended Layered-Borrowing Models

Appendix Table A13 reports the full set of coefficients for layered-borrowing models (Table 6 in main text), including all risk-factor controls and demographic variables.

10.8 A.8 Bootstrap Confidence Intervals and Inference Robustness

Appendix Table A15 reports household-level bootstrap confidence intervals (1,000 replications) for key coefficients in the main models. Bootstrap standard errors are uniformly smaller than HC3 standard errors, reflecting positive within-household correlation of residuals. The use of HC3 rather than bootstrap inference is therefore conservative.

10.9 A.9 Multicollinearity and Diagnostics

Variance inflation factors (VIFs) for all variables in the full specification are reported in Appendix Table A16. No variable has a VIF exceeding 3.0, indicating that multicollinearity is not a substantive concern.

Appendix Table A17 reports leverage and influence diagnostics for the main models. The DFFITS and Cook's D statistics identify several high-leverage observations but no points with implausible influence (Cook's $D > 0.01$).

10.10 A.10 Linear Probability Model Diagnostic Checks

Appendix Figure A1 plots the distribution of fitted values from the main welfare-selection model. Fitted probabilities range from -0.042 to 0.689, with approximately 8 percent of fitted values lying outside $[0,1]$. This is within acceptable bounds for an LPM with $n=6,131$ and a binary outcome with prevalence of 12.3%.

Appendix Figure A2 plots residuals against fitted values, showing approximately constant variance (no strong pattern).

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